

Total No. of Questions : 4]

SEAT No. :

PE-558

[Total No. of Pages : 2

[6578]-31

**S.E. (Information Technology) (Insem.)**

**BASICS OF COMPUTER NETWORK**

**(2019 Pattern) (Semester - III) (214445)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) *Attempt Q1 or Q2 & Q3 or Q4*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

**Q1) a)** Draw & Explain the Components of Communication System. [5]

b) Explain the four levels of addressing scheme in TCP/IP. [5]

c) Compare OSI & TCP/IP Model. [5]

OR

**Q2) a)** Explain the following layers in TCP/IP Model. [5]

i) Internet Layer

ii) Transport Layer

b) List & Discuss the Data Link Layer (DLL) functionality. [5]

c) State & Explain Shannon - Hartley theorem. [5]

**P.T.O.**

- Q3)** a) Define Error & Explain Different types of error during transmission. [5]
- b) Explain Checksum error detection technique with example. [5]
- c) Calculate the CRC if the data to be sent is 100100. The generator polynomial is  $G(x) = x^3 + x^2 + 1$ . [5]

OR

- Q4)** a) Define & Explain types of framing in Data Link Layer. [5]
- b) Explain stop-and-wait protocol. [5]
- c) Explain the terms [5]
- i) Piggybacking
  - ii) Parity check

